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In [1]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose

# Create sample website traffic data
np.random.seed(42)

dates = pd.date_range(start='2025-01-01', periods=120, freq='D')
page_views = (
    500
    + np.arange(120) * 2
    + 50 * np.sin(2 * np.pi * np.arange(120) / 7)
    + np.random.normal(0, 20, 120)
)

# Add a few traffic spikes
page_views[40] += 200
page_views[85] += 250

data = pd.DataFrame({
    'Date': dates,
    'Page_Views': page_views
})

data.set_index('Date', inplace=True)

# Display the dataset
print("Website Traffic Data:")
print(data.head())

# 1. Time Series Decomposition
decomposition = seasonal_decompose(
    data['Page_Views'],
    model='additive',
    period=7
)

decomposition.plot()
plt.suptitle("Time Series Decomposition", y=1.02)
plt.tight_layout()
plt.show()

# 2. Moving Average
data['Page_Views_MA'] = data['Page_Views'].rolling(
    window=7
).mean()

data[['Page_Views', 'Page_Views_MA']].plot(
    figsize=(10, 5)
)
plt.title("Page Views with Moving Average")
plt.xlabel("Date")
plt.ylabel("Page Views")
```

```
plt.show()

# 3. Anomaly Detection using Z-score
mean = data['Page_Views'].mean()
std = data['Page_Views'].std()

data['Page_Views_Z'] = (
    data['Page_Views'] - mean
) / std

anomalies = data[data['Page_Views_Z'].abs() > 3]

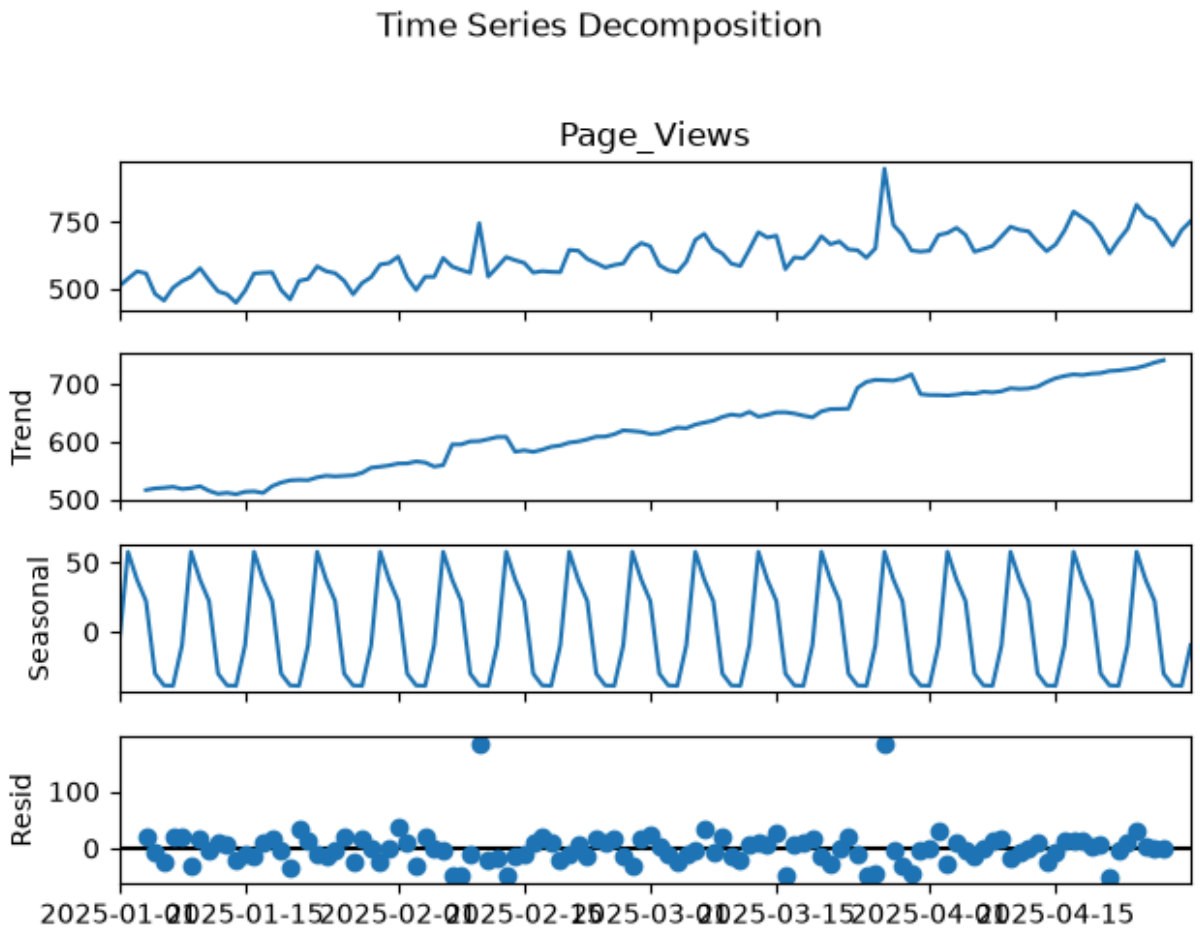
print("\nDetected Anomalies:")
print(anomalies[['Page_Views', 'Page_Views_Z']])

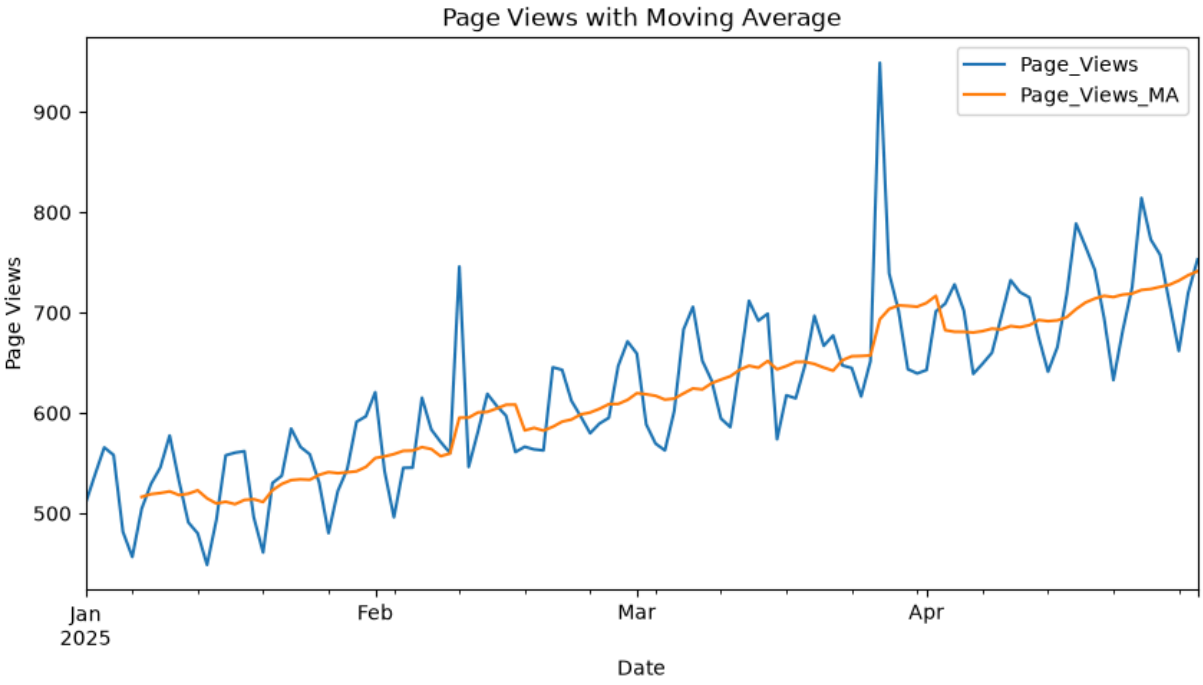
# Plot anomalies
plt.figure(figsize=(10, 5))
plt.plot(
    data.index,
    data['Page_Views'],
    label='Page Views'
)

plt.scatter(
    anomalies.index,
    anomalies['Page_Views'],
    color='red',
    label='Anomalies'
)

plt.title("Website Traffic Anomaly Detection")
plt.xlabel("Date")
plt.ylabel("Page Views")
plt.legend()
plt.show()
```

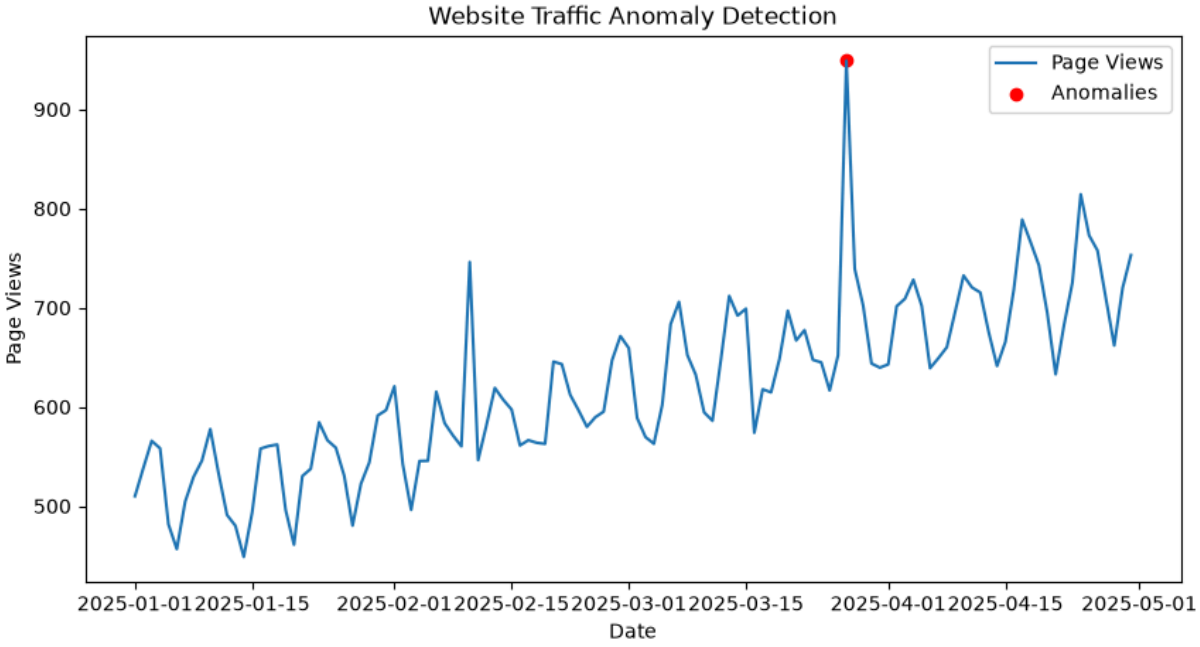
```
Website Traffic Data:
            Page_Views
Date
2025-01-01  509.934283
2025-01-02  538.326288
2025-01-03  565.700166
2025-01-04  558.154784
2025-01-05  481.622746
```





Detected Anomalies:

Date	Page_Views	Page_Views_Z
2025-03-27	949.056433	3.823627


Website Traffic Anomaly Detection
Page Views
Anomalies
900
800
700
600
500
Page Views
2025-01-01
2025-01-15
2025-02-01
2025-02-15
2025-03-01
2025-03-15
2025-04-01
2025-04-15
2025-05-01
Date

In [ ]: